Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We respond below to R1–ZQwd, R2–jAyD, and R3–Fcwd. The new evidence directly addresses the requested progress baselines, optimization ambiguity, mask scope, component ablations, and evaluation protocol. We will revise the paper to narrow the claim to an online, tactile-supervised short-horizon contact prior that reduces pre-contact stall.

[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning? We thank the reviewers and AC for highlighting this valuable comparison. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. Table I shows that VPCE consistently improves contact-prediction AUC/F1 over both alternatives. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

TABLE I
CONTACT-PREDICTION AUC/F1.

Task	Phase	Prog.	VPCE
SWIPE	$.670/.000$	$.853/.597$	$\mathbf{.963}/\mathbf{.759}$
WIPE	$.666/.000$	$.743/.406$	$\mathbf{.885}/\mathbf{.705}$
INSERT	$.684/.000$	$.808/.539$	$\mathbf{.946}/\mathbf{.735}$

Policy-level SWIPE: Phase-DP, Progress-DP, and VPCE-DP achieve $12/20$, $14/20$, and $28/30$. Thus, VPCE improves execution by predicting future contact states online with a state-dependent 15-step horizon.

[R1, R3] Does the teacher transfer, and how much target tactile data is required? The source VPCE teacher is trained on five OmniVitac task families—Cutting, Grasping, Wiping, Adjustment, and Assembly. After trajectory balancing, the source dataset contains 353 complete tactile trajectories, split by trajectory into 301 training and 52 held-out trajectories with no overlapping frames. For downstream target-domain training, we use $54/59/61$ tactile trajectories for WIPE/SWIPE/INSERT, respectively. On trajectory-disjoint held-out data, the target-domain teachers obtain AUC/F1 $.912/.731$, $.960/.774$, and $.940/.725$. Thus the source teacher covers diverse contact regimes, while approximately 60 target trajectories suffice to generalize to unseen trajectories in each downstream domain.

[R2] Are the learned loss weights well posed? The original design emphasizes sparse but control-critical contact-transition samples relative to abundant free-space samples. However, R2 is correct: because γ and δ are optimized through the same weighted denoising loss, the model can reduce the objective by lowering the weights without improving denoising. We therefore replace the learned term with the fixed schedule $w(\bar{c}) = \text{clip}(1 + \alpha\bar{c}, 1, 1 + \alpha)$, where α is a preselected hyperparameter and receives no gradients. This preserves the contact-aware emphasis while removing the optimization shortcut.

[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth? The reviewer is correct that the paper's notation and original post-encoder routing conflated proprioception, gripper width, and tactile features. Although input-level dropout targeted tactile keys, the post-encoder mask could zero the entire low-dimensional tail. We corrected the routing to always retain robot state and gripper width and mask only tactile/contact features. This controlled variant obtains $13/20$ on SWIPE and produces more coherent free-space approach motion. Action-chunk prediction and receding-horizon execution provide additional temporal continuity. We will revise Eq. (9) to distinguish f_{prop} from f_{tac}.

[AC, R3] Which components matter, and why retain gating? We thank the Area Chair and R3 and will move the compact component ablation from the appendix into the main text. The submitted SWIPE results are full $9/10$, no horizon $2/10$, horizon only $0/10$, no tactile masking $0/10$, no gating $9/10$, and no loss weighting $8/10$. These results identify future-contact horizon conditioning and contact-adaptive tactile masking as the core success mechanisms, with loss weighting serving as secondary training emphasis. The identical $9/10$ success rates show that binary success alone does not capture the role of gating. As shown qualitatively in the supplementary video, the gated policy enters contact more directly and maintains alignment, whereas the no-gating policy requires repeated corrective inference/action chunks before establishing aligned contact. Thus gating improves contact-entry trajectory quality by adaptively rebalancing visual and low-dimensional feedback, even when the final success rate is unchanged. We will add this qualitative comparison and the feature-weight traces to the main text.

[R2] Could operator-held props assist the policy? The reader and foam were held only to keep them at a predefined height and orientation within the robot's reachable range. After rollout start, the holder did not track the end effector, reposition the object, or assist alignment; all methods used the same placement, success criterion, and 60-s timeout. Crucially, this same protocol did not substantially rescue non-VPCE baselines on SWIPE: DP-V and DP-VT achieved only $1/20$ and $2/20$, versus $17/20$ and $18/20$ for VPCE-DP-V and VPCE-DP. Thus holding the reader at a reachable pose alone does not explain the method gap; with the predicted contact horizon, the policy committed smoothly through the final approach into contact and manipulation, directly addressing the pre-contact stall targeted by this work. We further expanded VPCE-DP to 30 trials per task: SWIPE $28/30$ (93.3%), WIPE $22/30$ (73.3%), and INSERT $18/30$ (60.0%).

This setup places the interaction within the wrist view and the 15-step contact horizon; VPCE is intended to regulate imminent contact rather than plan a long-range approach. For longer-horizon tasks, future work can add a wider or additional camera and a coarse approach stage before wrist-based VPCE regulates the final approach-to-contact transition.

[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled? VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame z-force proxy. A threshold of 0.095 is used for the source, SWIPE, and INSERT data; WIPE uses 0.0507 due to its different sensor scale. Each threshold is selected once from the training force-change distribution and then fixed. A $0.6\times$–$1.4\times$ threshold sweep changes the 15-step positive-label rate smoothly (SWIPE $.296 \rightarrow$ .217, WIPE $.582 \rightarrow$ .379, INSERT $.390 \rightarrow$ .290), rather than creating labels only at one tuned value. Contacts with changing normals can instead use $\|\Delta\mathbf{F}\|_2$, local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning? We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.